



PROOF OF CONCEPT (PoC) VALIDATION REPORT

Machine Hawk – Edge AI Predictive Maintenance Platform

1. PROJECT OVERVIEW

Title: Machine Hawk – Edge AI Predictive Maintenance Platform

Objective: Machine Hawk is an end-to-end predictive maintenance system that combines embedded sensing, Edge AI, Industrial IoT communication, backend analytics, and a web dashboard to detect abnormal machine behaviour before equipment failure.

Core Technologies: ESP32 microcontroller · MPU6050 inertial measurement unit · DHT22 temperature & humidity sensor · Edge Impulse (on-device inference) · MQTT messaging · Python backend · React web dashboard

2. PROOF OF IMPLEMENTATION

The following subsystems have been implemented and verified as part of the prototype:

Subsystem	Status
ESP32 Firmware	Completed
Sensor Integration	Completed
Edge AI Deployment	Completed
MQTT Communication	Operational
Backend Analytics	Operational
Web Dashboard	Operational
Real-Time Alerts	Operational
Health Score Generation	Operational
AI Recommendations	Operational

3. WORKING SYSTEM

The end-to-end data path of the prototype is as follows:

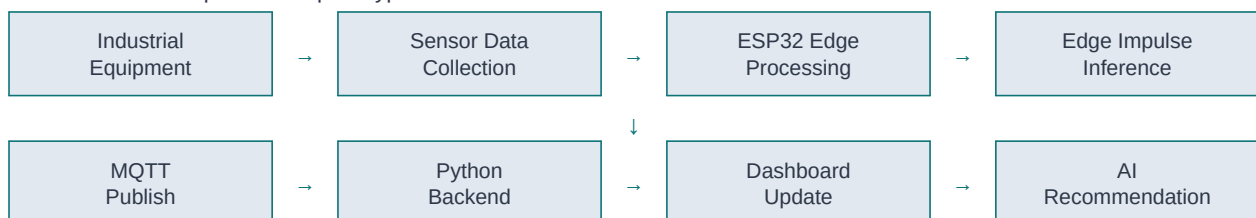


Figure 1. End-to-end system data flow from industrial equipment to AI recommendation.

4. PROTOTYPE VALIDATION

The images below are captured from the working prototype and constitute direct evidence of hardware, firmware, communication, and application-layer functionality.

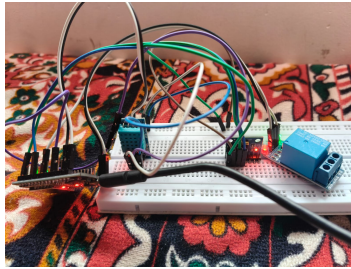


Figure 2. Hardware prototype – ESP32 development board interfaced with DHT temperature/humidity sensor, current-sensing modules, and relay on a solderless breadboard. Power and signal LEDs confirm active operation.

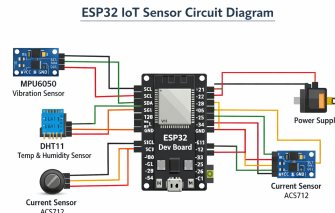


Figure 3. ESP32 IoT sensor circuit diagram showing MPU6050 (vibration), DHT (temperature & humidity), ACS712 current sensors, and power-supply connections to the ESP32 development board.

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INFO: 127.0.0.1:53403 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
INFO: 127.0.0.1:53406 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
2026-07-28 11:33:38,332 [INFO] machine_hawk.mqtt: Ingested telemetry update for machine 1 on topic machine-hawk/telemetry/1
INFO: 127.0.0.1:53409 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
INFO: 127.0.0.1:53412 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
2026-07-28 11:33:40,892 [INFO] machine_hawk.mqtt: Ingested telemetry update for machine 1 on topic machine-hawk/telemetry/1
INFO: 127.0.0.1:53415 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
INFO: 127.0.0.1:53418 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
INFO: 127.0.0.1:53421 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
2026-07-28 11:33:43,349 [INFO] machine_hawk.mqtt: Ingested telemetry update for machine 1 on topic machine-hawk/telemetry/1
INFO: 127.0.0.1:53424 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
INFO: 127.0.0.1:53427 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
2026-07-28 11:33:45,801 [INFO] machine_hawk.mqtt: Ingested telemetry update for machine 1 on topic machine-hawk/telemetry/1
INFO: 127.0.0.1:53430 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
INFO: 127.0.0.1:53433 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
INFO: 127.0.0.1:53436 - "GET /api/alerts?limit=50 HTTP/1.1" 200 OK
2026-07-28 11:33:48,368 [INFO] machine_hawk.mqtt: Ingested telemetry update for machine 1 on topic machine-hawk/telemetry/1
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Figure 4. Backend log output confirming continuous ingestion of telemetry on topic machine-hawk/telemetry/1 together with successful GET /api/alerts responses (HTTP 200).

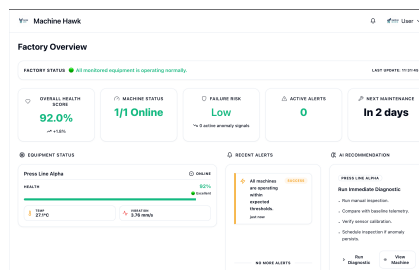


Figure 5. Live Factory Overview dashboard. Machine “Press Line Alpha” reports ONLINE status, health score 92.0 %, temperature 27.1 °C, vibration 3.76 mm/s, zero active alerts, and an AI recommendation panel.

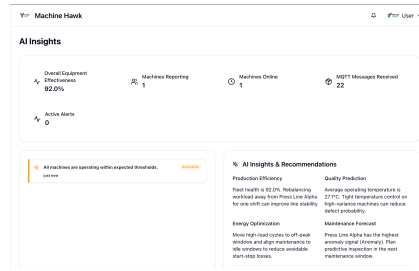


Figure 6. AI Insights view showing Overall Equipment Effectiveness (92.0 %), one machine reporting/online, 22 MQTT messages received, zero active alerts, and generated recommendations for production efficiency, quality prediction, energy optimisation, and maintenance forecast.

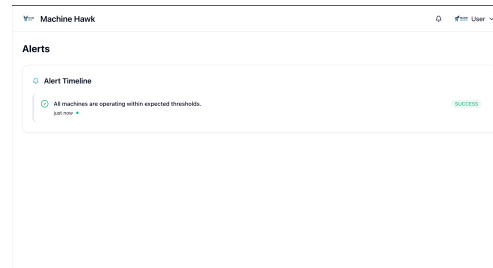


Figure 7. Real-time Alert Timeline indicating that all monitored machines are operating within expected thresholds (SUCCESS status).

5. VALIDATION

Each functional stage of the prototype was exercised and observed to operate correctly:

Validation Test	Result
Sensor Data Acquisition	PASS
Edge AI Inference	PASS
MQTT Communication	PASS
Backend Processing	PASS
Dashboard Visualization	PASS
Real-Time Alerts	PASS
System Integration	PASS

6. CONCLUSION

The Proof of Concept has been successfully demonstrated. The prototype performs:

- Real-time sensor acquisition (temperature, humidity, vibration, current)
- Embedded AI inference on the ESP32 via Edge Impulse
- MQTT-based telemetry publication and backend ingestion
- Backend processing and health-score computation
- Live dashboard visualisation of machine state and alerts
- Generation of predictive-maintenance recommendations

Machine Hawk therefore demonstrates the technical feasibility of an Edge AI-powered predictive maintenance platform suitable for Industrial IoT and Industry 4.0 applications.

PoC Status: All planned subsystems of the Machine Hawk prototype have been implemented, integrated, and observed in operation. The evidence presented in this report (hardware photographs, circuit documentation, MQTT logs, and live dashboard screenshots) constitutes verification of functional end-to-end behaviour.